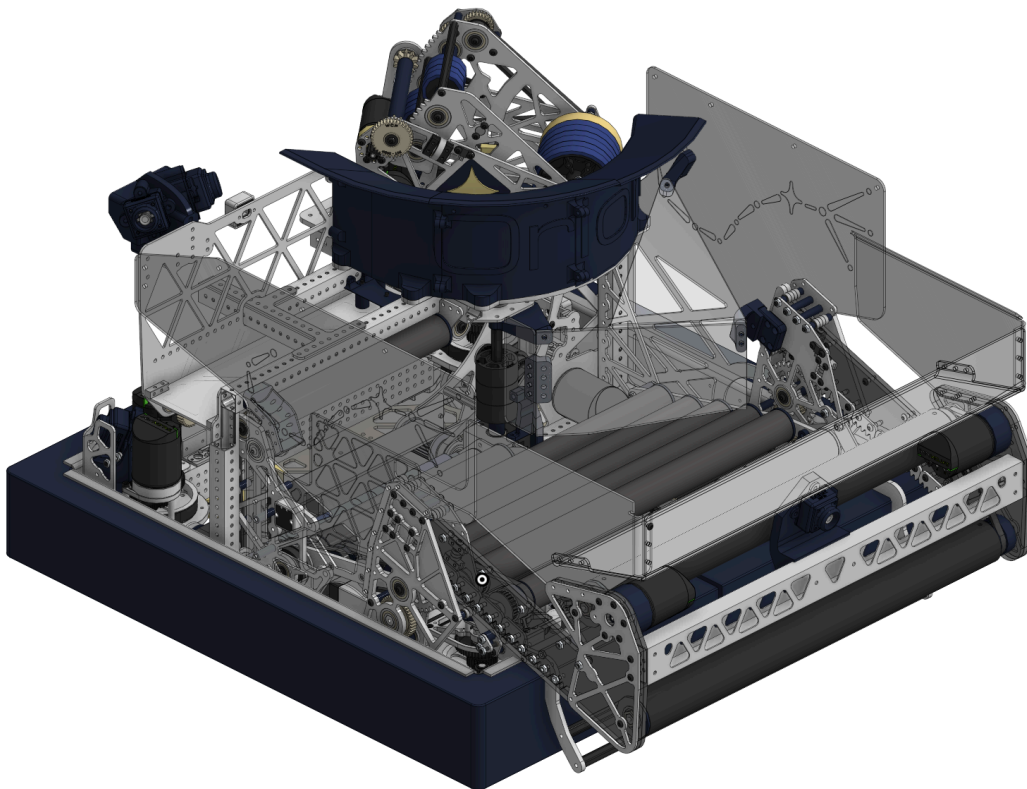


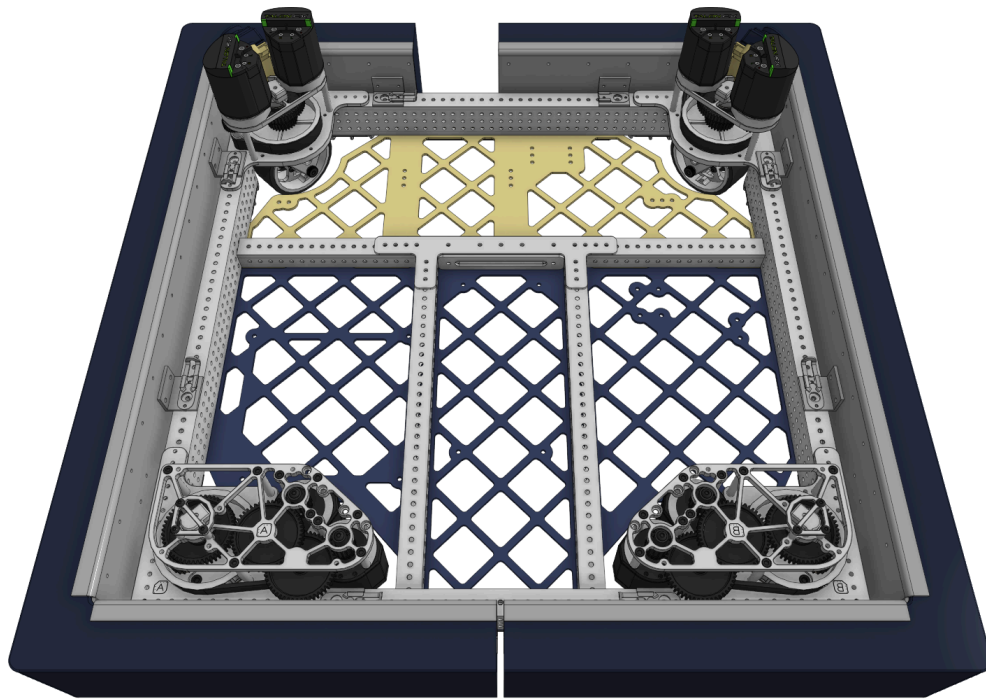


2026 Tech Binder



Orion

Drivetrain



The Drivetrain lets Orion maneuver around the field, quickly reaching the neutral zone to pick up fuel and driving back to the alliance zone to shoot. This swerve drivetrain is designed to reach high speeds for rapid navigation between zones, withstand hard impacts when under defense, and easily slip over the bump.

Chassis & Bumpers

- Square frame with 27" x 27" outer dimensions
- 2" x 1" box tubing on exterior edges supported by inner 1" x 1" rectangular tubing
- 1/8" 6061 lightened aluminum bellypan adds shear support and holds up electronics
- Bumper foam is backed with high strength 1/8" 5053 bent aluminum frame

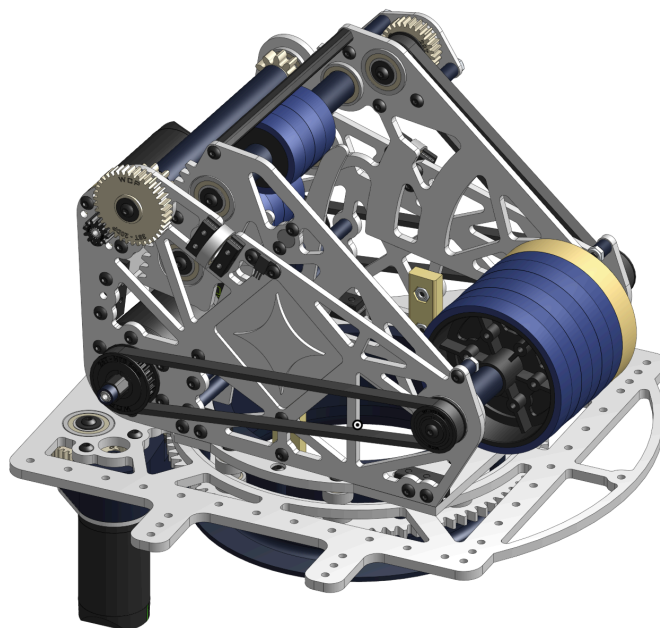
MK4c & MK4i Swerve Modules

- High drive motor gear ratio selected for greater top speed
- Multiple modules designs utilized for increased hopper capacity
- Motions are precisely synchronized using motion profiling, feedforward control, and PID

Low-Friction Skids

- Four low-friction acetal skids underneath each outer frame tube prevent high-centering when crossing the bump

Shooter



The Shooter propels high volumes of fuel into the hub from any location within the alliance zone, and the rotating turret lets Orion shoot while moving to avoid defense and intake fuel in parallel.

Rollers

- Fuel is propelled out of the shooter at high speeds by three sets of rollers
- One large custom roller 3D printed out of TPU (not shown) applies most of the speed
- 2" stealth wheels on the actuated hood reduce backspin, which was predicted to increase shot variability using a custom-built simulation

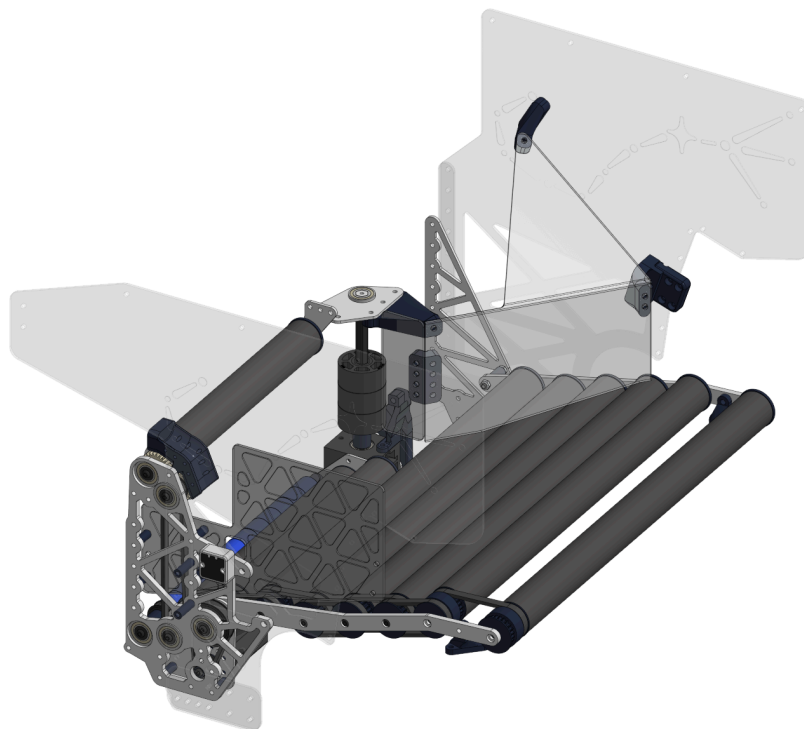
Hood

- 2" stealth wheels pivot coaxially around main roller to adjust shot angle
- 25° range enables passing from across the full field and shooting from close to the hub
- Hall effect sensor detecting magnet in the pivoting hood zeroes the encoder automatically
- Bearings perpendicular to the hood maintain smooth motion and dissipate vibrations

Turret

- Rotates the shooter to enable shooting fuel in any direction
- Custom 10DP gear with a hole for fuel allows precise movement with $<2^\circ$ of backlash
- Drag chain connects wires from main robot to rotating shooter
 - Stows automatically using a constant force spring into a long channel
 - Length of channel and flippable connection allows for 520° of turret rotation
 - Latching WAGO 221 connectors at either end enable quick replacement

Hopper



The hopper stores fuel within the robot and contains a moving roller floor to shuttle fuel down into the indexer for shooting.

Fuel Storage

- Can store up to 43 fuel while still going under the trench
- Covered with a net to prevent fuel from falling out when passing over the bump
- Two CANRange time-of-flight IR distance sensors determine when the hopper is full or empty

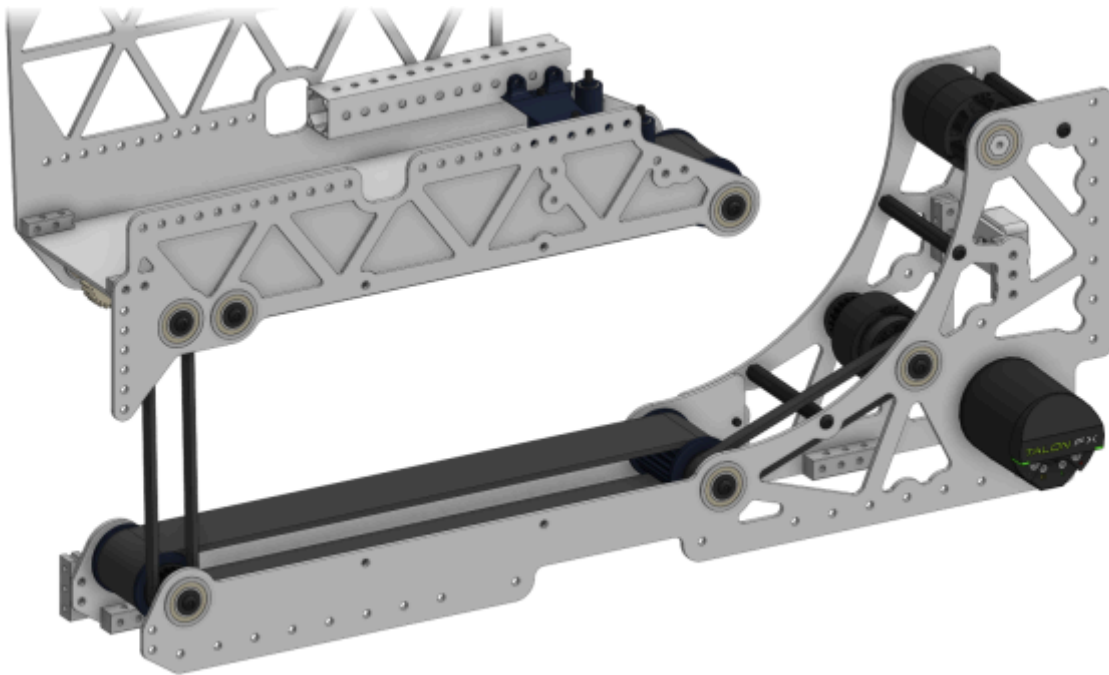
Roller Floor

- Moves fuel towards the indexer by spinning a set of silicone-wrapped polycarbonate rollers under fuel
- Vertical roller reduces chance of jamming as fuel passes through the indexer
- Uncovered polycarbonate roller gently nudges fuel into the indexer from above, without pushing hard enough to delay fuel already into the indexer from going into the shooter
- Pivots up for easy access to electronics on belly pan

Passive Funnel

- Polycarbonate funnel at the corner of the hopper guides fuel towards the indexer entrance
- Thumb screws enable quick removal for access to electronics underneath

Indexer



The indexer quickly transfers a two-ball wide stream of fuel from the hopper to a single-file stream into the shooter.

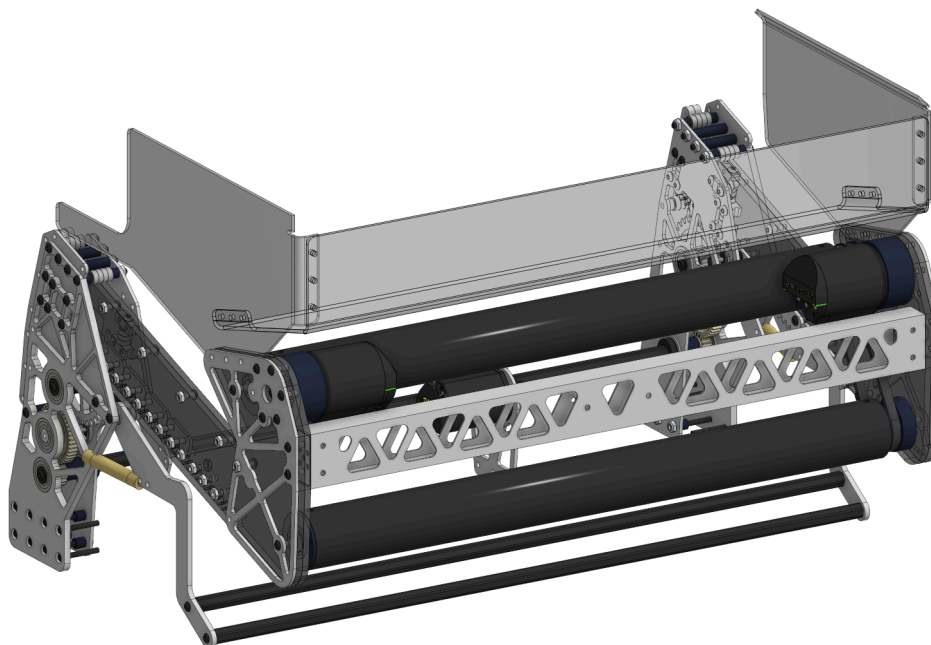
Queue

- The “queue” allows fuel from the roller floor to enter from the side and routes them using conveyor belts along the queue towards the kicker
- Uses very wide belts to reduce risk of belts inverting due to forces from fuel entering the queue
- Cutout in upper plate enables two balls to easily roll into the queue at once

Kicker

- The kicker accelerates balls from the queue to a consistent speed and propels them up into the shooter
- Compliant wheel and belt on both sides of exit point ensure fuel exits the kicker with minimal rotation that would cause variability in shooting
- A CANrange time-of-flight IR distance sensor behind the kicker detects when fuel is loaded into the queue, enabling automatic loading of fuel from the hopper into the queue to reduce shooting startup delay

Intake



Orion's intake efficiently scoops fuel off the ground and lifts it into the hopper. Our fourth iteration of the intake, this final design can withstand hard collisions from defense and continues intaking fuel at near optimal speed even when the robot is moving quickly through dense masses of fuel.

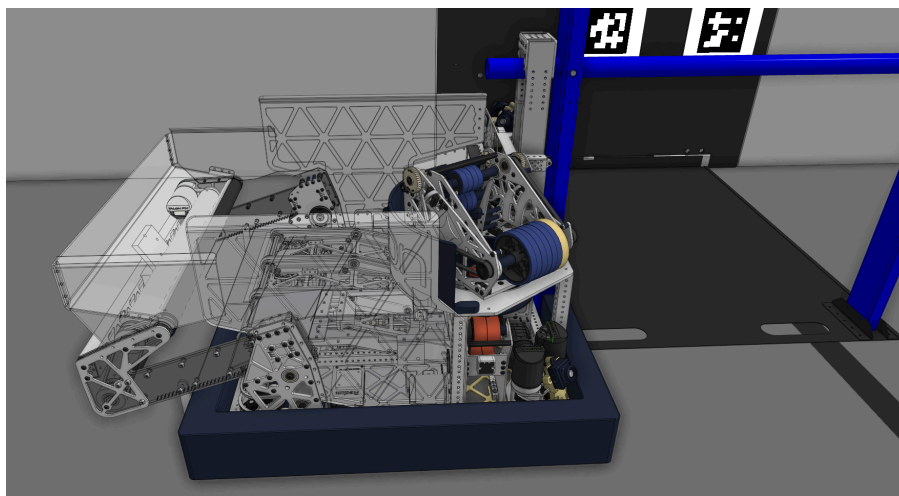
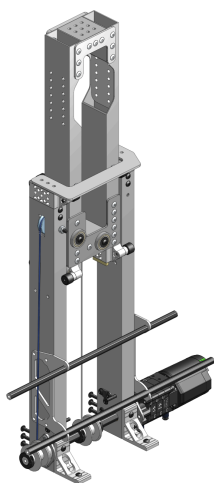
Rollers

- Top speed of 30-40 fuel per second
- Maintains high performance even when robot is moving fast due to fast surface speed
- The passively deploying kicker bar ensures that the two silicone-wrapped polycarbonate rollers maintain compression with fuel
- Rollers are powered by two Kraken X60s to enable higher speed and prevent overheating
- 2" x 1" rectangular aluminum tube on the front takes the first hit during matches

Rack & Pinion Extension

- Linear slide extension design lets the intake passively retract when hit head-on
- Bearings on all side of rack eliminate nearly all extension friction
- Replaceable 1/8" 6061 aluminum teeth on rack, a large pinion gear, and all gear-driven powertrain prevents tooth damage and skipping during collisions
- Side plates are made of 1/2" polycarbonate for durability and flexibility
- Hall effect sensor detects a magnet embedded in the polycarbonate plates zeroes the extension automatically before matches

Climb



The now-removed Climb mechanism was a one-stage elevator that lifts the robot off the ground by grabbing the L1 bar. It was removed before Silicon Valley District Event because the traversal ranking point is unlikely to be scored due to other alliance members rarely having climb mechanisms.

Elevator

- One stage internally-rigged elevator driven by one Kraken X44 lifts and lowers hook
- Elevator is driven using low-stretch, high strength UHMWPE cord
- MAXPlanetary breaking stage allows the robot to stay off the ground without power
- The elevator is designed to be compatible with a more complex L3 climb design
- Hall effect sensor detects a magnet in the carriage to zero the elevator extension

Hook

- Hook uses the robot's weight to pinch the bar
- V shape ensures high friction between hook and L1 bar to keep the robot from falling off

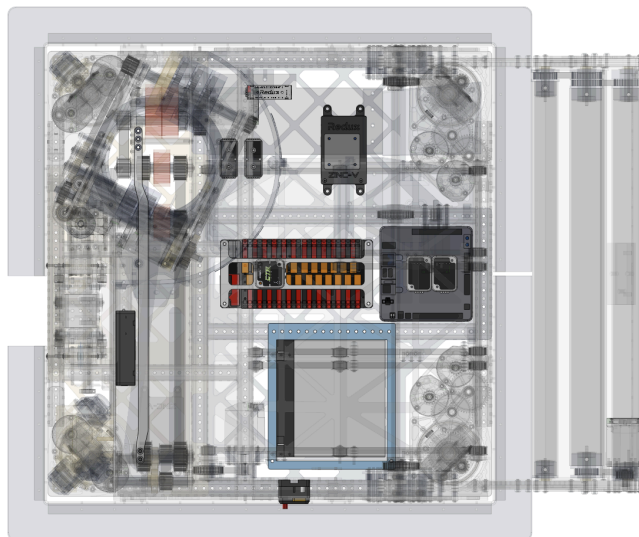
Bumper Cutout

- Bumper cutout in the back of the robot allows the hook to grab onto the bar without extending outside the frame perimeter

Load Distribution

- Elevator maintains 6" between upper and lower bearings when climbed to avoid bending
- The robot's triangular superstructure distributes the load of the robot's weight when climbing

Electronics



Orion's electrical system has been carefully designed to minimize the chances of damage during matches making the robot non-functional and make fixing problems between matches easy.

Power Distribution

- Power is supplied to fourteen Kraken X60 and four Kraken X44 motors from the PDP 2.0 through 40A breakers
- To lower robot weight and remove less reliable connectors, large power wires are directly soldered to many smaller wires instead transferring power through a Mini Power Module

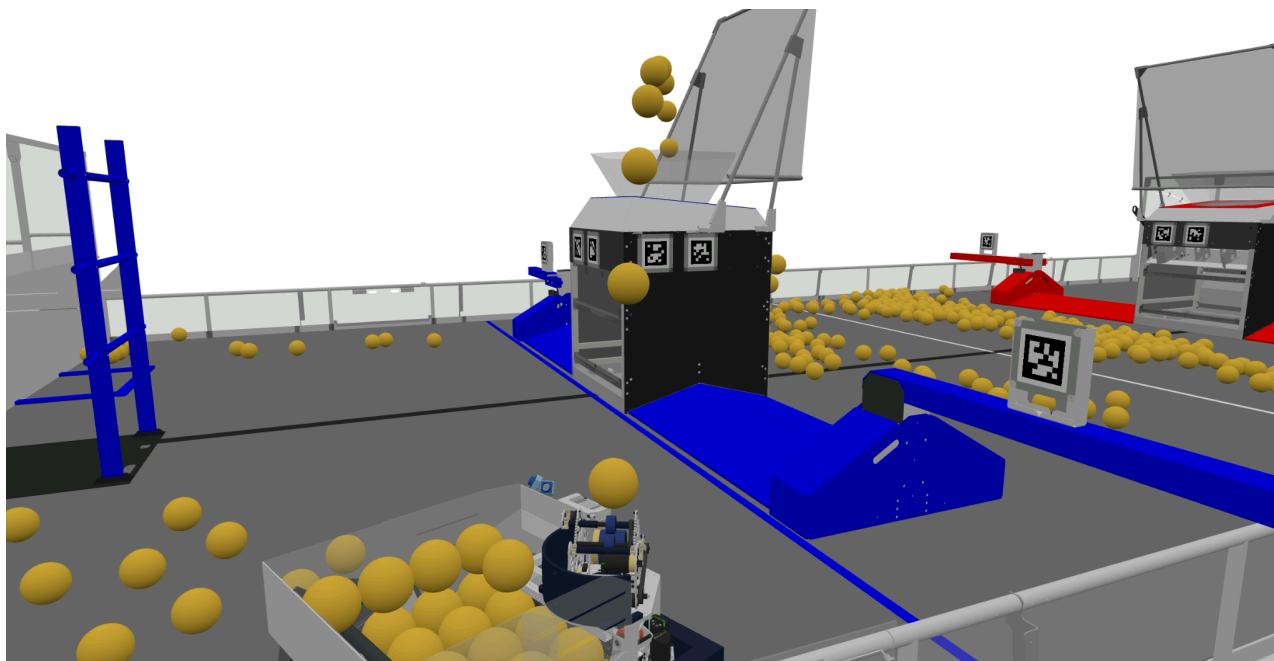
Reliability

- The number of connectors has been minimized to remove possible failure points
 - Large 10 AWG power wires have continuous runs directly from PDP 2.0 to motors
 - Small 22 AWG wire connections are soldered to prevent random disconnection
- All connections are carefully strain relieved using zip ties
- Main breaker is accessible but protected from bumps from fuel by a polycarbonate plate
- Separation of subsystems and drivetrain into two independent CAN FD buses keep the robot drivable even if wires in the turret or intake break
- Subsystem CAN bus is terminated inside the moving intake and turret, so some subsystems continue to function even if wires are damaged
- Hall effect sensors are used instead of limit switches for zeroing, so mechanism can't become non-functional because switches are bent or broken

Maintainability

- Pivoting belt floor allows wiring on the belly pan to be easily maintained
- The turret drag chain is connected with strain relieved WAGOs and soldered connections, allowing speedy replacement of the entire drag chain

Simulation



Complete, physically accurate 3D robot simulation allows us to test mechanism code, autonomous routines, controls changes, and automation in parallel with work on the real robot and other code.

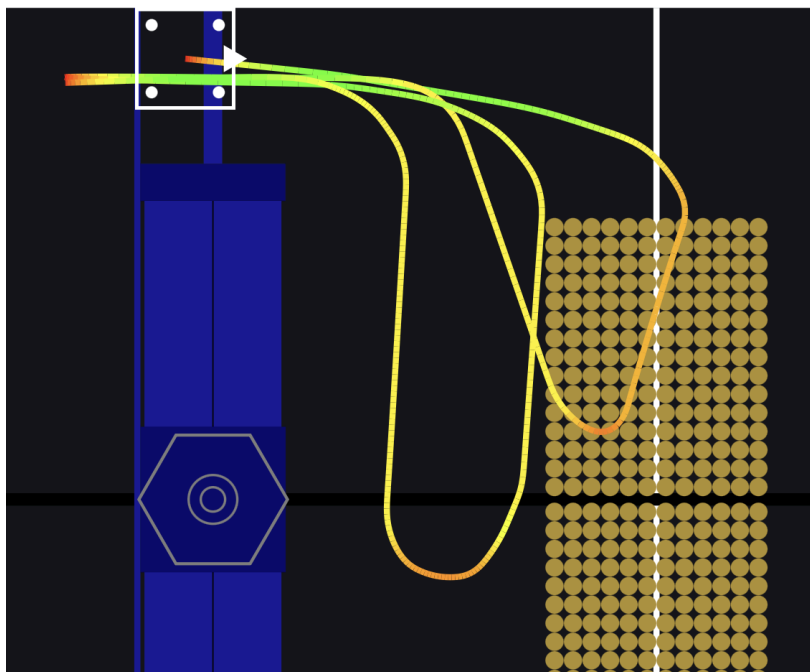
Combines Simulation Libraries and Team-Built Code

- Phoenix 6 CAN bus and motor controller logic simulation
- WPILib mechanism physics simulation, modified to accurately simulate unusual mechanisms, including the slanted linear intake extension
- MapleSim to simulate the drivetrain and fuel on the field
- 3D visualization using AdvantageScope for rendering
- Custom simulation for fuel in the robot and distance sensor data

Reduced Code Testing Time

- This resulted in large sections of code working perfectly the first time they were deployed to the robot — because they'd already been tested thoroughly in simulation
- Autonomous routines are fully tested and iterated upon in simulation, meaning that only final optimization demands precious robot time

Autonomous



The autonomous software controls the robot during the autonomous phase of the match, aims the turret and hood, preemptively loads fuel into the kicker, and prevents the hood from breaking when passing through the trench.

Autonomous Shooting

- Orion can calculate an accurate shooting trajectory from any position on the field to any other position on the field, either for passing back to the alliance zone or shooting into the hub
- Shooting calculations use empirically collected hood angle and flywheel speed to shoot from stationary positions
- Flight time data is used to compensate for robot motion when shooting on the move

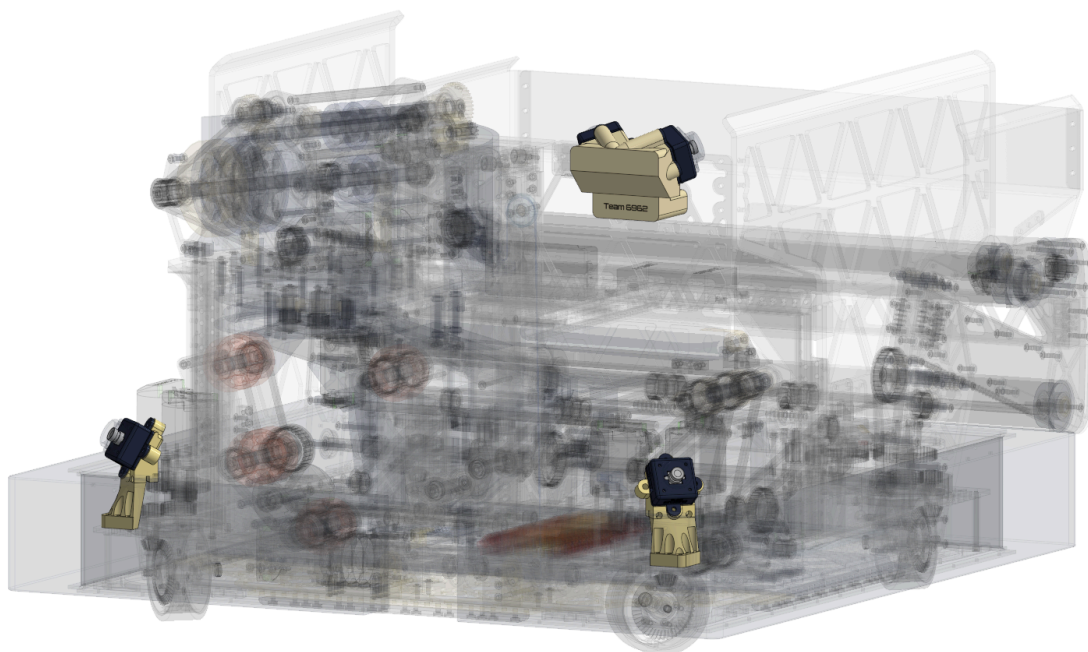
Refined Autonomous Routines

- During the autonomous phase of the match, Orion typically drives to the neutral zone, intakes fuel, and then returns to the alliance zone to shoot into the hub
- Autonomous paths are optimized with Choreo to reach the neutral zone as fast as possible to steal fuel from the center line

Mechanism Automation

- Shooter hood automatically lowers when near a trench to avoid hitting the roof
- Roller floor preemptively loads fuel into the queue when fuel is detected in the hopper to reduce shooter warm up times

Vision



The vision system uses 4 cameras to detect AprilTags, locating the robot's position on the field and enabling it to automatically aim the shooter.

Sensors

- The four cameras detecting AprilTags for robot localization are placed in strategic locations that enable accurate localization anywhere in the alliance zone
- Encoders in swerve modules and Pigeon 2 gyroscope collect relative motion data at higher frequencies to reduce latency and update location when AprilTags are out of view

Algorithms

- A multi-tag PhotonVision pose estimator collects data from the AprilTags seen by each camera, with a fallback single-tag lowest-ambiguity pose estimation method
- Relative motion data from odometry and gyroscope is fused using a Kalman filter with AprilTag detections from cameras to provide accurate absolute robot position and heading data
- To make recalibrating odometry for tread wear during competition easy, the effective radius of the wheels is determined automatically by rotating the robot in place and comparing motor encoder data with gyroscope angles

Controls

Orion's software controls interpret commands from the driver or autonomous logic and power motors to make mechanisms quickly and precisely follow those commands.

Custom Swerve Drive Control Stack

- Orion's high-performance swerve drive control stack is fully built by Antares, providing valuable learning to students not accessible when using off-the-shelf swerve libraries
- Provides most of the same features of standard libraries and more
- Highly command-based API makes creating and improving autonomous routines fast and accessible for newer code subteam members
- Low-level swerve control loop runs in a thread at 100 Hz instead of the typical 50 Hz, decreasing drift when rotating while translating
- Motion profiling and feedforward control for all swerve motors avoids drift present when using other swerve libraries to control Orion's drivetrain, which has multiple different module designs

Fast & Precise Mechanism Control

- All mechanisms where precision of motion is desired are controlled with motion profiling, feedforward control, and PID, utilizing more advanced control algorithms that result in more precise and time-optimal motion than PID on its own
- The feedforward equations for all mechanisms were derived from motor physics laws, to accurately predict control constants based on values in CAD without using valuable robot time for tuning
- All motors have stator and supply current limits tuned to avoid brownouts during matches or breaking mechanisms while maximizing scoring rate